

Temporal Multi-Omics (TMO)

Complete Mathematical Description

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1 Notation

- N : number of cells
- G : number of genes
- P : number of ATAC peaks
- $K = 50$: number of latent components per modality
- $d_{\text{model}} = 64$: token embedding dimension
- $H = 4$: number of attention heads
- $L = 2$: number of transformer layers
- $\tau \in [0, 1]$: pseudotime
- $Z_{\text{RNA}} \in \mathbb{R}^{N \times K}$: RNA latent components
- $Z_{\text{ATAC}} \in \mathbb{R}^{N \times K}$: ATAC latent components
- $a_i(\tau), r_i(\tau)$: standardised ATAC and RNA component scores
- $\Delta\tau$: signed regulatory lag (pseudotime units)
- $\Delta\tau_{\text{max}} = 0.5$: maximum absolute lag
- $\sigma_{\text{min}}^2 = 0.001$: minimum window width
- $\lambda_{\text{recon}} = 1.0, \lambda_{\text{lag}} = 0.1$: loss weights
- $\sigma_0^2 = 0.01$: prior variance (absorbed in loss scaling)

2 Data Preprocessing

2.1 Gene-level ATAC matrix

Let $X_{\text{peak}} \in \mathbb{R}_{\geq 0}^{N \times P}$ be the peak-by-cell count matrix. Given a binary peak-to-gene annotation matrix $M \in \{0, 1\}^{P \times G}$, where $M_{p,g} = 1$ if peak p is assigned to gene g , the gene-level ATAC matrix is

$$A_{\text{gene}} = X_{\text{peak}} M \in \mathbb{R}_{\geq 0}^{N \times G}. \quad (1)$$

A_{gene} is stored as a sparse CSR matrix in `.obsbm['ATAC_gene']`. Raw RNA counts remain in `.X`.

2.2 Pseudotime

RNA counts are normalised to 10^4 total reads per cell, $\log(1 + x)$ -transformed, and the top 2000 highly variable genes are selected. After PCA (30 components) and k -NN graph ($k = 15$), diffusion maps are computed (`scanpy.tl.diffmap`). The first diffusion component is min-max scaled to $[0, 1]$ and taken as pseudotime τ .

If an early marker gene (e.g. *Pax6*, *CD34*) is provided, its expression vector x_m is correlated with the initial pseudotime τ_0 :

$$r = \text{Pearson}(x_m, \tau_0).$$

If $r < 0$, the pseudotime is reversed: $\tau = 1 - \tau_0$, and a flag `pseudotime_reversed` is stored. In all downstream steps, if the flag is true, the sign of every $\Delta\tau$ is flipped so that

positive $\Delta\tau$ always means ATAC leads RNA (priming).

3 Latent Representations

3.1 RNA components

$$Z_{\text{RNA}} = \text{PCA}_{50}(\log(1 + X_{\text{RNA}})) \in \mathbb{R}^{N \times K}.$$

3.2 ATAC components

$$X_{\text{tfidf}} = A_{\text{gene}} \text{diag}(\log \frac{N}{\mathbf{d}}), \quad d_j = \#\{c : A_{\text{gene}}(c, j) > 0\}. \quad (2)$$

Each row (cell) of X_{tfidf} is ℓ_2 -normalised. Then truncated SVD yields

$$Z_{\text{ATAC}} = \text{TruncatedSVD}_{50}(X_{\text{tfidf}}) \in \mathbb{R}^{N \times K}.$$

A cell is represented by a sequence of $2K = 100$ tokens: the first K tokens are the ATAC components, the remaining K tokens are the RNA components.

4 Ground-Truth Regulatory Lag from CCF

For each matched component pair (i, i) , we estimate a signed, time-resolved lag surface along pseudotime.

4.1 Interpolation and cross-correlation

The pseudotime axis is non-uniform. Within a sliding window of half-width $w = 0.1$ centred at $\tau_0 \in \{0.1, 0.2, \dots, 0.9\}$, the ATAC and RNA series are linearly interpolated onto a dense regular grid of 200 points. The cross-correlation is computed with `scipy.signal.correlate` in full mode:

$$\text{CCF}_i(\tau_0, \ell) = \frac{1}{|I|} \sum_{t \in I} a_i(\tau_t) r_i(\tau_t + \ell),$$

where $I = \{t : |\tau_t - \tau_0| \leq w\}$ and ℓ is the lag in pseudotime units. The peak lag is

$$\Delta\tau_i(\tau_0) = \arg \max_{\ell \in [-0.3, 0.3]} \text{CCF}_i(\tau_0, \ell).$$

4.2 Gaussian process smoothing

The raw surface $\Delta\tau_i(\tau)$ is smoothed with a Gaussian process (Matern-3/2 kernel, length-scale bounds $[10^{-2}, 0.5]$, noise-level bounds $[10^{-4}, 10^0]$). The training target for component i is the smoothed value at the last window:

$$\Delta\tau_i^{\text{target}} = \Delta\tau_i^{\text{smooth}}(\tau = 0.9). \quad (3)$$

If the pseudotime was reversed, the whole surface is multiplied by -1 , so that the target has the correct biological sign.

5 Tokenisation

Each component’s scalar value v is projected to d_{model} via a learned linear layer $\text{proj} : \mathbb{R}^1 \rightarrow \mathbb{R}^{d_{\text{model}}}$. The token embedding for a component with latent value v , modality m ($m = 0$ for RNA, $m = 1$ for ATAC), and pseudotime τ is

$$\mathbf{x} = \text{proj}(v) + \mathbf{e}_{\text{mod}}(m) + \mathbf{e}_{\tau}(\tau), \quad (4)$$

where

- $\mathbf{e}_{\text{mod}} \in \mathbb{R}^{d_{\text{model}}}$ is a learned modality embedding,
- $\mathbf{e}_{\tau}(\tau)$ is the output of a 2-layer MLP $\mathbb{R}^1 \rightarrow \mathbb{R}^{128} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^{d_{\text{model}}}$ (ReLU activations).

The same pseudotime embedding is shared by all tokens of a cell. The sequence of $2K$ tokens is formed as

$$\mathbf{X} = [\mathbf{x}_{\text{ATAC},1}, \dots, \mathbf{x}_{\text{ATAC},K}, \mathbf{x}_{\text{RNA},1}, \dots, \mathbf{x}_{\text{RNA},K}] \in \mathbb{R}^{2K \times d_{\text{model}}}.$$

6 Transformer Encoder

The encoder consists of $L = 2$ identical TMO blocks. Each block contains:

6.1 Within-modality self-attention

Standard multi-head self-attention (no bias) is applied separately to the ATAC tokens and to the RNA tokens.

6.2 Asymmetric cross-modal attention

Cross-attention from one modality to the other is biased by the predicted lag. The mechanism is described in section 8.

6.3 Feed-forward network

A 2-layer MLP with expansion factor 4:

$$\text{FFN}(\mathbf{x}) = \mathbf{W}_2 \text{GELU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2,$$

followed by dropout and residual connections.

Each sub-layer is preceded by LayerNorm and followed by a residual connection.

7 Lag and Width Prediction

The cell embedding $\mathbf{z}_c$ is obtained by mean-pooling the ATAC token outputs of the first (unbiased) pass:

$$\mathbf{z}_c = \frac{1}{K} \sum_{i=1}^K \mathbf{h}_{\text{ATAC},i}^{(1)} \in \mathbb{R}^{d_{\text{model}}}.$$

For each component index i (separate for ATAC and RNA), a learnable component embedding $\mathbf{e}_{\text{comp}}(i) \in \mathbb{R}^{d_{\text{model}}}$ is concatenated with $\mathbf{z}_c$ and fed to two shared MLPs.

7.1 LagMLP

$$\mathbf{h}_1 = \text{LayerNorm}([\mathbf{z}_c; \mathbf{e}_{\text{comp}}(i)]), \quad (5)$$

$$\mathbf{h}_2 = \text{GELU}(\text{Linear}_{2d_{\text{model}} \rightarrow d_{\text{model}}}(\mathbf{h}_1)), \quad (6)$$

$$\mathbf{h}_3 = \text{Dropout}_{0.1}(\mathbf{h}_2), \quad (7)$$

$$\mathbf{h}_4 = \text{GELU}(\text{Linear}_{d_{\text{model}} \rightarrow d_{\text{model}}/2}(\mathbf{h}_3)), \quad (8)$$

$$\mathbf{h}_5 = \text{Dropout}_{0.1}(\mathbf{h}_4), \quad (9)$$

$$\hat{\Delta\tau}_{\text{raw}} = \text{Linear}_{d_{\text{model}}/2 \rightarrow 1}(\mathbf{h}_5), \quad (10)$$

$$\hat{\Delta\tau} = \Delta\tau_{\text{max}} \cdot \tanh(\hat{\Delta\tau}_{\text{raw}}). \quad (11)$$

7.2 WidthMLP

Identical architecture to LagMLP, except the final activation:

$$\hat{\sigma}^2 = \sigma_{\min}^2 + \text{Softplus}(\text{MLP}_{\text{Width}}(\mathbf{z}_c, \mathbf{e}_{\text{comp}}(i))).$$

The predictors are applied to all K ATAC and K RNA indices, yielding

$$\hat{\Delta\tau}_{\text{ATAC}}, \hat{\sigma}_{\text{ATAC}}^2 \in \mathbb{R}^{N \times K}, \quad \hat{\Delta\tau}_{\text{RNA}}, \hat{\sigma}_{\text{RNA}}^2 \in \mathbb{R}^{N \times K}.$$

8 Asymmetric Attention Bias

In the second forward pass, cross-attention logits are additively biased.

8.1 ATAC $\rightarrow$ RNA

For an ATAC token at pseudotime τ_i attending to an RNA token at pseudotime τ_j ,

$$b_{ij} = - \frac{(\tau_j - \tau_i - \hat{\Delta\tau}_{\text{RNA},j})^2}{2\hat{\sigma}_{\text{RNA},j}^2}. \quad (12)$$

8.2 RNA $\rightarrow$ ATAC

For the reverse direction,

$$b_{ji}^{\text{rev}} = - \frac{(\tau_i - \tau_j + \hat{\Delta\tau}_{\text{ATAC},i})^2}{2\hat{\sigma}_{\text{ATAC},i}^2}. \quad (13)$$

The biases are added to the scaled dot-product attention scores before softmax, making the attention mechanism explicitly asymmetric. Within-modality attention (ATAC-ATAC, RNA-RNA) uses unbiased attention.

8.3 Windowed sparse attention (optional)

For large datasets, attention can be restricted to token pairs satisfying

$$|\tau_j - \tau_i - \hat{\Delta\tau}| \leq 3\hat{\sigma}.$$

This reduces the complexity from $\mathcal{O}(N^2L^2)$ to $\mathcal{O}(NLW)$ where W is the constant window size.

9 Decoder

After the second pass, the K RNA token outputs are linearly projected to reconstruct the RNA latent components:

$$\hat{\mathbf{R}} = \text{Linear}_{d_{\text{model}} \rightarrow 1}(\mathbf{H}_{\text{RNA}}^{(2)}) \in \mathbb{R}^{N \times K}.$$

10 Training Loss

The model is trained end-to-end on the full dataset with

$$\mathcal{L}_{\text{total}} = \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{lag}} \mathcal{L}_{\text{lag}}, \quad (14)$$

where $\lambda_{\text{recon}} = 1.0$, $\lambda_{\text{lag}} = 0.1$.

10.1 Reconstruction loss

$$\mathcal{L}_{\text{recon}} = \frac{1}{NK} \sum_{c=1}^N \sum_{k=1}^K (\hat{\mathbf{R}}_{c,k} - Z_{\text{RNA},c,k})^2. \quad (15)$$

10.2 Lag consistency loss

For each ATAC component i , the predicted lag is averaged over cells: $\bar{\Delta\tau}_i = \frac{1}{N} \sum_c \hat{\Delta\tau}_{\text{ATAC}}(c, i)$. The lag loss is

$$\mathcal{L}_{\text{lag}} = \frac{1}{K} \sum_{i=1}^K (\bar{\Delta\tau}_i - \Delta\tau_i^{\text{target}})^2, \quad (16)$$

where $\Delta\tau_i^{\text{target}}$ is defined in (3).

10.3 Symmetric baseline

A symmetric baseline is trained with the identical architecture but with `use_bias = False` (no attention bias) and $\lambda_{\text{lag}} = 0$ (no lag loss). The baseline therefore only minimizes $\mathcal{L}_{\text{recon}}$.

10.4 Optimization

Adam optimizer ($\eta = 10^{-3}$, weight decay = 0.01) with a stepwise learning-rate scheduler ($\gamma = 0.5$ at epochs 20 and 40). Batch size 32, 50 epochs. Every 5 epochs the Lag Concordance Score (LCS) is computed on the full dataset (in-set) and the best checkpoint is saved.

11 Lag Concordance Score (LCS)

The LCS is the Spearman rank correlation between the per-component mean predicted lags and the CCF-derived targets:

$$\text{LCS} = \rho_{\text{Spearman}}\left(\{\bar{\Delta\tau}_i\}_{i=1}^K, \{\Delta\tau_i^{\text{target}}\}_{i=1}^K\right).$$

NaN targets are excluded. $\text{LCS} > 0.9$ indicates that the model correctly orders the components by their regulatory lag. The LCS is reported as an in-set diagnostic; independent biological validation is provided by ChIP-seq and Perturb-seq experiments.

12 Cell-State Ablation Experiment

To verify that the cell-state-dependent nature of TMO’s lag predictions originates from the LagMLP’s access to the cell embedding, we performed an ablation in which the cell embedding is replaced by the zero vector before being passed to the LagMLP and WidthMLP.

12.1 Mathematical description

In the ablated model, the LagMLP and WidthMLP inputs become

$$\mathbf{h}_1 = \text{LayerNorm}([\mathbf{0}; \mathbf{e}_{\text{comp}}(i)]), \quad (17)$$

$$\hat{\Delta\tau}_{\text{ablated}} = \Delta\tau_{\text{max}} \cdot \tanh(\text{Linear}(\dots)), \quad (18)$$

$$\hat{\sigma}_{\text{ablated}}^2 = \sigma_{\text{min}}^2 + \text{Softplus}(\dots), \quad (19)$$

where $\mathbf{0}$ is the d_{model} -dimensional zero vector and the rest of the architecture remains unchanged. The ablation therefore forces the predictor to output a fixed scalar per component, independent of the cell’s state.

12.2 Quantitative results

The ablated model was trained on all four datasets. The Lag Concordance Score dropped consistently (e.g., from 0.9984 to 0.7136 in mouse kidney; from 0.9990 to 0.9112 in human PBMC; see Supplementary Table ST4). To measure the loss of temporal dynamics, we computed the linear slope of the predicted $\Delta\tau$ against pseudotime for each component. The full model exhibited a broad distribution of slopes (mean absolute slope 0.0026–0.1114 across tissues), whereas the ablated model’s slopes were sharply concentrated near zero (mean absolute slope 0.0044–0.0092). A two-sample Kolmogorov–Smirnov test confirmed that the slope distributions differ significantly in all four tissues (KS $p < 0.05$ in every case and $p < 10^{-18}$ in three of the four).

These results demonstrate that cell-state information is indispensable for the dynamic, pseudotime-dependent regulatory lags that TMO learns, and that the observed temporal patterns are not artefacts of the CCF target or the transformer architecture.

13 ChIP-seq Validation

For a transcription factor with top-500 bound genes, component-level CCF lags $\Delta\tau^{\text{CCF}} \in \mathbb{R}^K$ are projected to gene-level lags via the PCA loadings $\mathbf{L} \in \mathbb{R}^{G \times K}$:

$$\text{gene_lag} = \tilde{\mathbf{L}}^\top \Delta\tau^{\text{CCF}},$$

where $\tilde{\mathbf{L}}$ is obtained by row-normalising $|\mathbf{L}|$ so that each row sums to 1.

The full set of non-target genes (optionally restricted to those within ± 0.5 log-expression units of the target-gene median) is used as background. A two-sided Mann-Whitney U test compares the $\Delta\tau$ values of target and background genes. This test is deterministic (no random subsampling) and independent of the model predictions.

14 Perturb-seq Validation

For a knockout (e.g. SMARCB1), the asymmetric model is trained on non-targeting control cells. The per-component absolute lag shift $\delta\Delta\tau_i = |\bar{\Delta\tau}_i^{\text{pert}} - \bar{\Delta\tau}_i^{\text{ctrl}}|$ is compared between components associated with known target genes (identified via highest absolute PCA loading) and all other components using a one-sided Mann-Whitney U test.